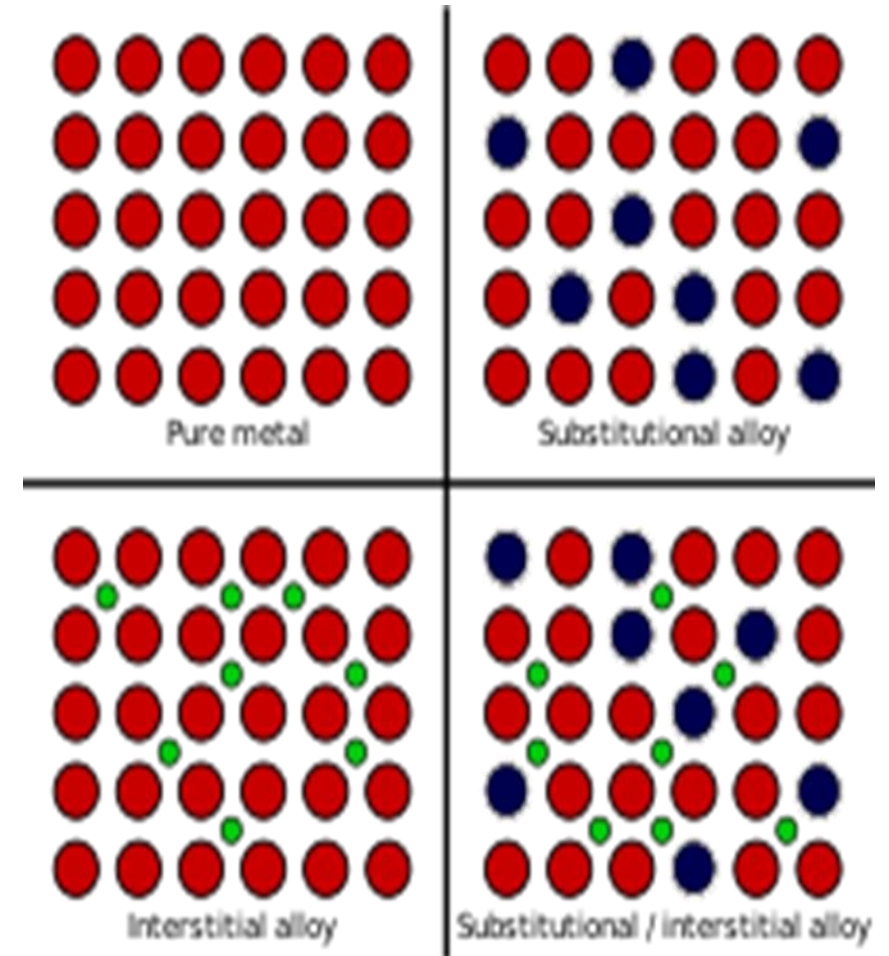


ME23-205-0305
METALLURGY & MATERIAL
SCIENCE
Module 4

Alloys

- Alloys are metallic compounds made up of one metal and one or more metal or non-metal elements.
- Examples of common alloys:
 - Steel: A combination of iron (metal) and carbon (non-metal)
 - Bronze: A combination of copper (metal) and tin (metal)
 - Brass: A mixture of copper (metal) and zinc (metal)
- Individual pure metals may possess useful properties such as good electrical conductivity, high strength, and hardness, or heat and corrosion resistance.
- Commercial metal alloys attempt to combine these beneficial properties in order to create metals more useful for particular applications than any of their component elements.
- Steel, for example, requires the right combination of carbon and iron (about 99% iron and 1% carbon) in order to produce a metal that is stronger, lighter, and more workable than pure iron.



Stainless steels

- Stainless steel, also known as inox, corrosion-resistant steel (CRES), and rustless steel, is an alloy of iron that is resistant to rusting and corrosion.
- It contains iron with chromium and other elements such as molybdenum, carbon, nickel and nitrogen depending on its specific use and cost.
- Stainless steel's resistance to corrosion results from the 10.5%, or more, chromium content which forms a passive film that can protect the material and self-heal in the presence of oxygen.
- The alloy's properties, such as luster and resistance to corrosion, are useful in many applications.
- Stainless steel can be rolled into sheets, plates, bars, wire, and tubing.

Austenitic Stainless Steels

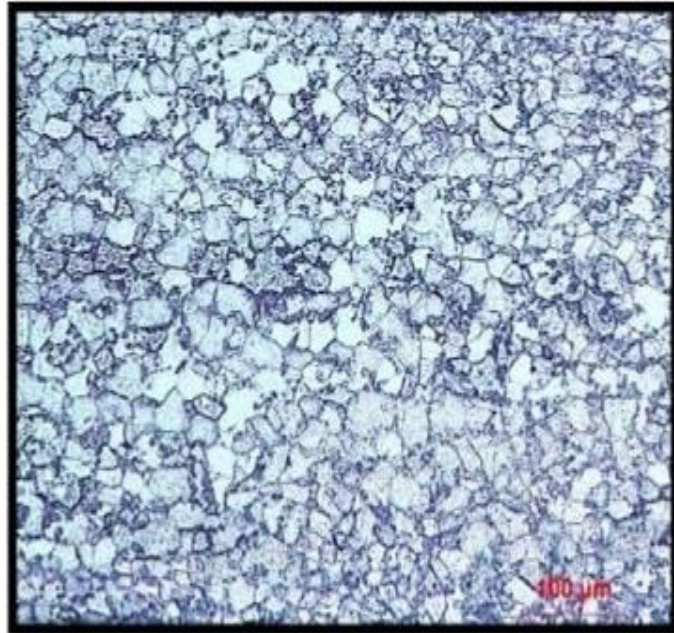
- These are the most frequently used types of stainless steels.
- Rich in chromium and nickel, austenitic stainless steel is renowned for its exceptional resistance to corrosion and excellent formability.
- Another common characteristic is that they tend to be non-magnetic—though they may become magnetic after cold working.
- Applications- kitchen equipment, industrial equipment, and chemical processing.

Ferritic Stainless Steels

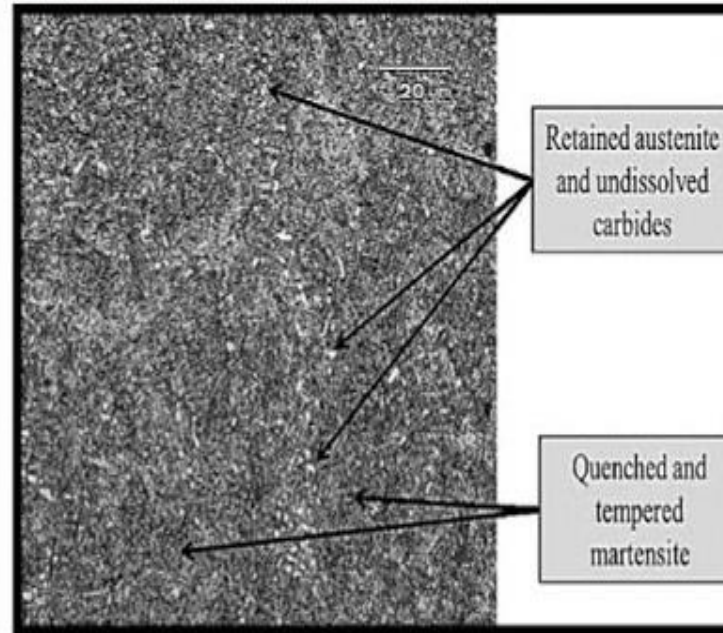
- The second most common form of stainless steel after austenitic alloys.
- Ferritic stainless steel has a unique microstructure, primarily composed of iron and chromium.
- Ferritic stainless steel is magnetic.
- These alloys can be hardened through cold working.
- They also tend to be less expensive because of their reduced nickel content.
- Applications- automotive parts, kitchen appliances, and even some architectural applications.

Stainless steels

Ferritic stainless steel



Martensitic stainless steel.



Austenitic stainless steels.

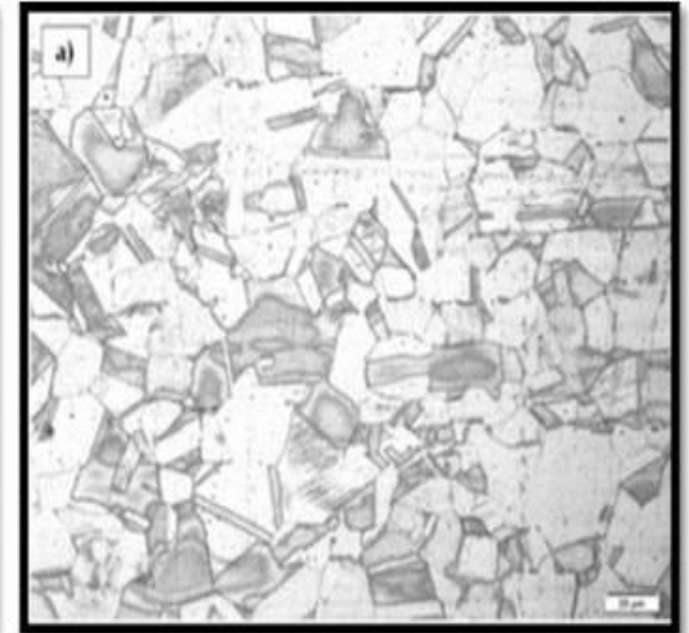
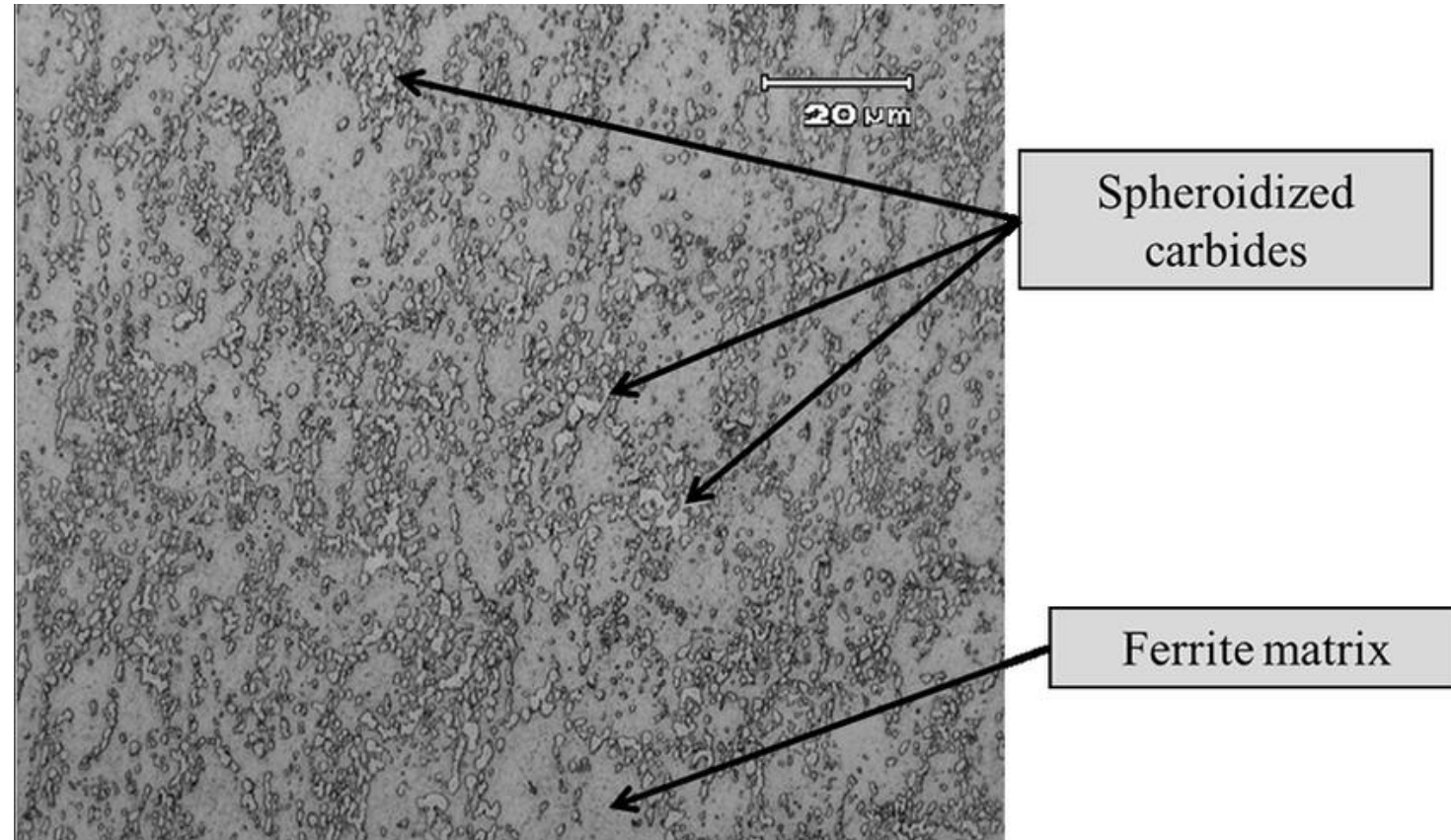


Table 1. Difference in the properties of ferritic and austenitic stainless steels.

Properties	Ferritic	Austenitic
Toughness	Moderate	Very high
Ductility	Moderate	Very high
Weldability	Limited	Good
Thermal expansion	Moderate	High
Stress corrosion cracking resistance	Very high	Low
Magnetic properties	Ferro magnetic	Non-magnetic

Martensitic Stainless Steels

- The least common category of stainless steel alloy.
- Their corrosion resistance tends to be lower than either ferritic or austenitic alloys, but they have a high hardness.
- Martensitic stainless steel alloys are often ideal for applications requiring extraordinarily high tensile strength and impact resistance.
- When said applications also require corrosion resistance, these alloys may be used with a protective polymer coating.



Duplex (Ferritic-Austenitic) Stainless Steels

- This category of stainless steel derives its name “duplex” from its composition; it is made of half austenite and half delta-ferrite.
- These stainless steel alloys have better corrosion resistance, especially against chloride pitting, and higher tensile strength than standard austenitic stainless steels.
- Due to their physical properties and chemical resistance, duplex stainless steels are used widely in the oil and gas industry for pipework systems or as pipelines and pressure vessels in the petrochemicals industry.

Precipitation-Hardening (PH) Stainless Steels

- PH stainless steels contain small amounts of copper, aluminum, phosphorus, or titanium, in addition to other stainless steel and nickel-based alloy components.
- PH stainless steels are solution treated and then quenched, and then subjected to a low-temperature heat treatment called ageing or age-hardening.
- The heat treatment process can be adjusted to achieve different levels of strength and toughness.
- PH stainless steels have a high strength-to-weight ratio.
- PH stainless steels are used in applications that require high strength and some corrosion resistance.
- This category of stainless steel is comprised of durable, corrosion-resistant alloys with exceptional strength.
- They can be treated to yield strengths three to four times that of standard austenitic stainless steel.
- They are most commonly used in the aerospace, nuclear, and oil and gas industries.

Tool Steels

- The name "tool steel" describes a range of carbon and alloy steels that are especially well-suited for use in the production of tools.
- These steels are distinguished by their hardness, resistance to wear, toughness, and resistance to softening at high temperatures.
- There are seven main types of tool steel: water-hardening, hot-work, cold-work, shock-resisting, mold steels (used to manufacture the molds, including cold-punching mold, hot forging die, die-casting mold etc.) high-speed steels, and special purpose tools steel.
- The carbon percentage in tool steels is typically in the range of 0.7 to 1.5 wt% carbon.
- Some tool steels can contain up to 2.1% carbon, while others contain less than 0.25 % carbon.
- Hardness, strength, and hardenability are enhanced with an increase in carbon concentration.

Tool Steels

- The three key properties associated with tool steel are: wear resistance, heat resistance, and toughness.
- Alloying elements are added to increase strength, wear resistance, hardness, and toughness.
- Tool steels are composed of various combinations of carbide-forming metals such as chromium, molybdenum, tungsten, and vanadium.
- These iron-based alloys have relatively high carbon content for strength and carbide formation.
- Substitutional solutes like nickel and cobalt are added for high-temperature strength.
- Carbide-forming metals like: chromium, molybdenum, tungsten, and vanadium are added for hardness and wear resistance.

Copper and its alloys

- Copper alloys are metal alloys that contain copper as a main component.

✓ Brass

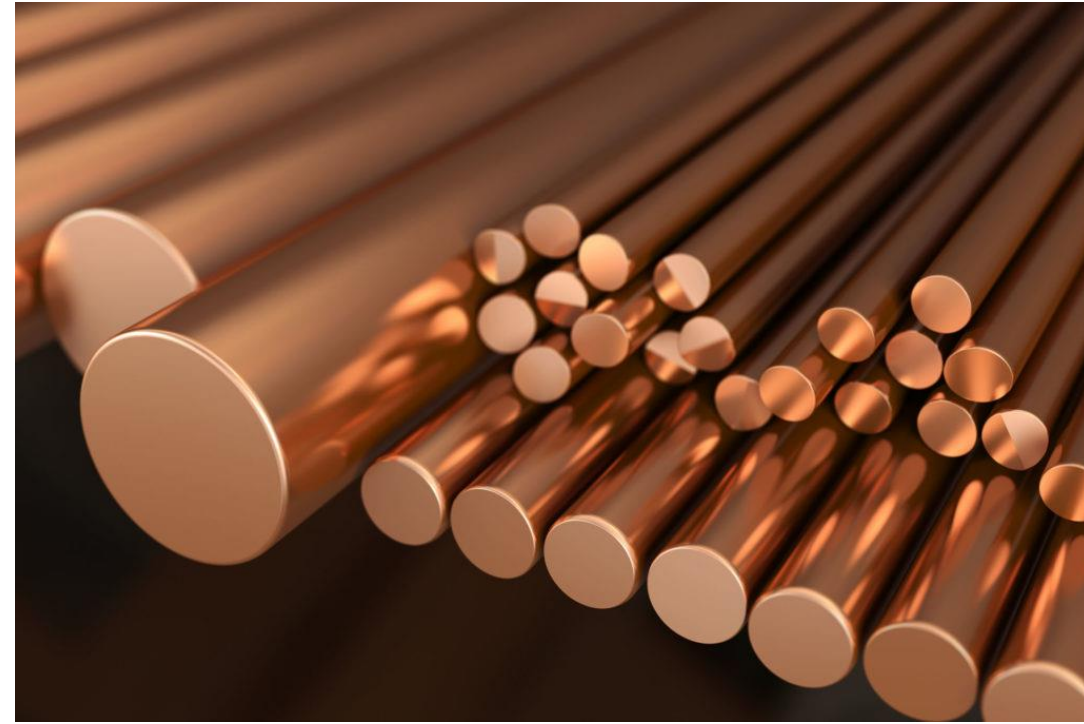
- A copper alloy made by adding zinc to copper.
- The properties of brass depend on the zinc content.

✓ Bronze

- A copper-tin alloy that's easy to weld, strong, and resistant to corrosion and wear.
- It's used to make heating accessories, building metal wares, and decorative parts.

✓ Copper powder

- A golden material made of copper alloy and a small amount of other metals, such as zinc, aluminum, and tin. It's used in decorative coatings to replace gold foil.



Copper and its alloys

✓ Bearing alloys

- Copper-based alloys that are considered superior bearing alloys due to their properties and cost.
- They have a maximum operating temperature of 450°F / 230°C and a load capacity of 4,000 lbs. per square inch for those with the highest tin content.
- Copper alloys are used in mechanical watches because they are easy to shape through machining or cold-forming.
- Other elements that are often added to copper alloys include:
 - Silicon: Reduces the melting point and increases fluidity in castings
 - Aluminum: Improves strength and corrosion resistance, and adds a yellow-gold tone
 - Manganese: Improves casting properties and enhances the strength of copper
 - Phosphorus: Improves ductility, strength, and corrosion resistance, and deoxidizes copper

Aluminium and its alloys

- Aluminium is often combined with other elements, such as silicon, tin, manganese, or even copper, tin, and magnesium, depending on the desired application.
- With the right combination of elements, aluminium can acquire a lot more strength, and in some cases, it could even outperform steel.
- Aluminium alloys are light in weight and corrosion resistant
- Alloys bring the same benefits as pure aluminium, and they are also relatively cost-effective because they have a lower melting point.
- Applications- aerospace and automotive, building and construction, transportation, electrical, medical equipment and implants.
- [https://www.aalco.co.uk/datasheets/Aluminium-Alloy Introduction-to-Aluminium-and-its-alloys 9.ashx](https://www.aalco.co.uk/datasheets/Aluminium-Alloy%20Introduction-to-Aluminium-and-its-alloys%209.ashx)

Magnesium and its alloys

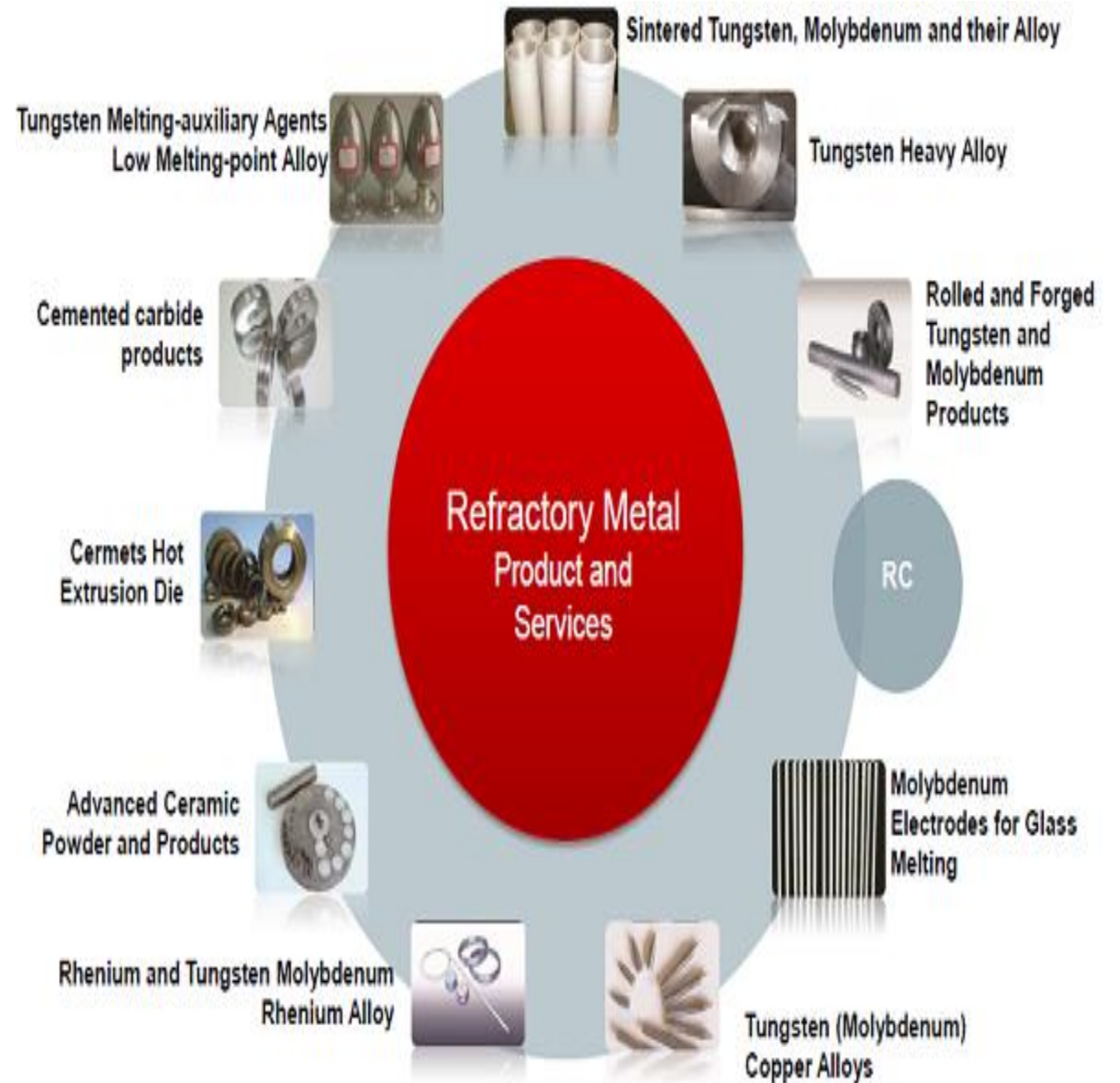
- Magnesium alloys are mixtures of magnesium (the lightest structural metal) with other metals (called an alloy), often aluminium, zinc, manganese, silicon, copper, rare earths and zirconium.
- Magnesium alloys have a hexagonal lattice structure, which affects the fundamental properties of these alloys.
- Magnesium alloys are considered as the lightest element than aluminium (33%), titanium (61%), and stainless steel (77%).
- Due to its lightweight and good specific strength, it is widely used in many food and medical industrial sectors as well as automobiles and aerospace.
- Now magnesium alloys are becoming an excellent alternative to steel and alloys of aluminium.
- The main factor that limits the application of magnesium alloys is their corrosive nature due to their high reactivity.
- Magnox (alloy) is 99% magnesium and 1% aluminium and is used in the cladding of fuel rods in magnox nuclear power reactors.
- Magnesium alloys are referred to by short codes (defined in ASTM B275) which denote approximate chemical compositions by weight. For example, AS41 has 4% aluminium and 1% silicon; AZ81 is 7.5% aluminium and 0.7% zinc.
- Mag-Thor is the common name for a range of magnesium (Mg) alloys containing thorium (Th) that are used in aerospace engineering.

Titanium and its alloys

- Titanium alloys typically contain traces of aluminum, molybdenum, vanadium, niobium, tantalum, zirconium, manganese, iron, chromium, cobalt, nickel, and copper.
- Titanium and its alloys are known for their strength, corrosion resistance, and lightweight nature. They are used in a variety of applications, including the military, aviation, and biomedical industries:
- They are stronger than pure titanium, and are also more flexible and corrosion resistant.
- ✓ Beta titanium
 - This alloy is made from titanium, vanadium, and aluminum, and is used in eyeglasses frames. The high strength to modulus ratio of this alloy makes it elastic, which helps keep eyeglasses stable during intense activity.
- ✓ Biomedical titanium alloys
 - These alloys are used in biomedical implants because of their high strength, corrosion resistance, and biocompatibility.
- ✓ T40
 - This alloy is nearly pure titanium, with small amounts of iron and oxygen added. It is used in the aeronautic industry.
- ✓ Ti-6Al-7Nb
 - This alloy is made from titanium, aluminum, and niobium, and was developed to replace the well-known Ti-6Al-4V alloy for medical applications.
- Refer: <https://waykenrm.com/blogs/types-of-titanium-alloys/>

Refractory Metals And Alloys

- Refractory metals are a group of metallic elements that are highly resistant to heat and wear.
- It is generally accepted that tungsten, molybdenum, niobium, tantalum and rhenium best fit most definitions of refractory metals.
- High wear and corrosion resistance
- Very low coefficient of thermal expansion
- Due to cost and availability reasons, alloying elements are used selectively and in smaller proportions
- Refractory ceramics are used in furnaces, crucibles, molten metal carrying vessels.



Refractory Metals And Alloys

- Refractory materials can be grouped as:
 - ✓ Acid Refractories- made of silica, aluminium silica, alumina,etc
 - ✓ Basic Refractories- made of magnesia, bauxite,dolomite,etc
 - ✓ Neutral Refractories: made of chromite, graphite,etc

Base Metal Properties					Carbides Properties		
Material	Symbol	Density g/cm ³	Melting Point		Carbide Symbol	Melting Point	
			° C	° F		° C	° F
Titanium	Ti	4.50	1660	3020	TiC	2781	5038
Zirconium	Zr	6.49	1852	3366	ZrC	3540	6404
Hafnium	Hf	13.1	2270	4118	HfC	3930	7106
Vanadium	V	5.80	1890	3434	VC	2800	5072
Niobium	Nb	8.55	2468	4474	NbC	3390	6134
Tantalum	Ta	16.6	2996	5425	TaC	3980	7196
Chromium	Cr	7.19	1857	3375	Cr ₃ C ₂	1895	3443
Molybdenum	Mo	10.2	2617	4743	Mo ₂ C	2690	4874
Tungsten	W	19.3	3410	6170	WC	2780	5036
Rhenium	Re	21.0	3180	5756	Re	2505	4541

Important Refractory Materials & Applications

Material	Composition	Working Temp.	Applications
Graphite	Carbon	3000	Heating Rods
Thoria	ThO ₂	2750	Jet Nozzles
Magnesia	MgO	2000	Basic steel making
Alumina	Al ₂ O ₃	1860	Thermocouple
Quartz	SiO ₂	1700	Alloy steel making
Fireclay	40-60% Al ₂ O ₃ 60-40% SiO ₂	1300	Furnace lining

Superalloy

- A superalloy is a metallic alloy which can be used at unusually high homologous temperatures, often in excess of 0.8 T_m
- Amazing property of superalloys: they become stronger at higher temperature :
 - Able to maintain high strengths
 - Good corrosion and oxidation resistance (Cr, Al)
 - Good resistance to creep and rupture
 - Good fatigue behaviour

▪ 3 main classes of superalloys:

- **Ni – Base**

- These are the most common type of superalloy and are often made with nickel, aluminum, and/or titanium.
- They also contain chromium, iron, cobalt, and other elements.
- Use : Jet engines and turbochargers because of their ability to retain their strength and stiffness at high temperatures

- **Fe – Base**

- These are cheaper than other superalloys and are known for their wear resistance. They are often made with iron and nickel or chromium.
- Iron-based superalloys are used in aircraft bearings and other applications that involve moving contact.

- **Co – Base**

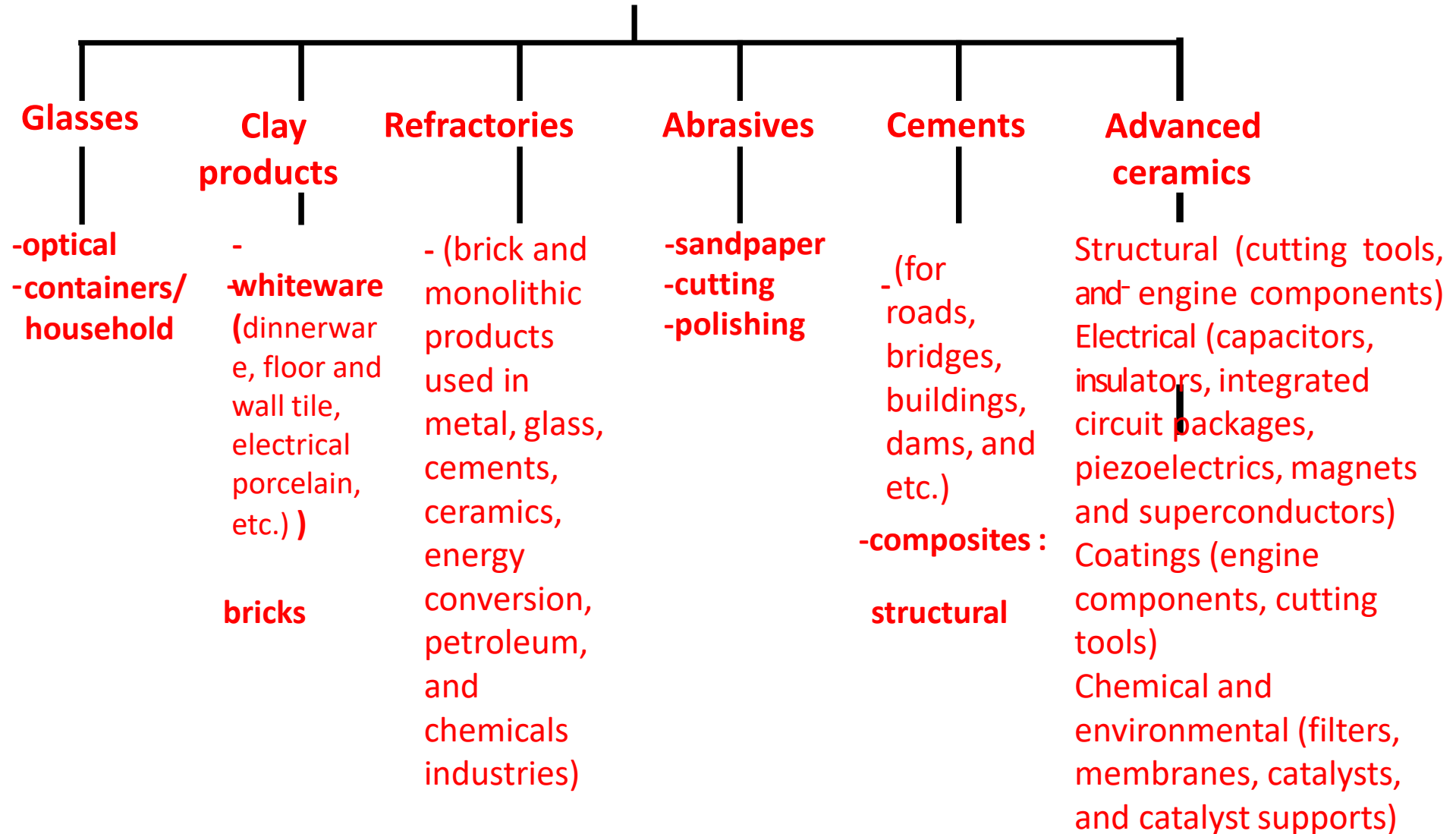
- Made with cobalt, chromium, and tungsten.
- They have a high fatigue life and are stronger than iron- and nickel-based superalloys at high temperatures.
- They are also highly resistant to corrosion

Ceramics

- **Ceramics are compounds between metallic and nonmetallic elements; they are most frequently oxides, nitrides, and carbides.**
- **The wide range of materials that falls within this classification includes ceramics that are composed of clay minerals, cement, and glass.**
- **These materials are typically insulate electricity and heat, and are more resistant to high temperatures and harsh environments than metals and polymers.**
- **Ceramics are hard but very brittle.**



Classification Of Ceramics



Types of Clay

Earthenware

- most common type of clay
- low fire
- porous
- opaque, soft, and capable of being scratched when fired
- example: terracotta



Stoneware

- fired at high temperature
- nonporous
- difficult to scratch
- can be described as having a "dirty look"



Porcelain

- high fire clay
- wet very quickly
- considerably strong and hard
- vitrified (water cannot pass through)
- white
- difficult to work with
- translucent (light can go through)



Ceramics

- A ceramic has traditionally been defined as “an inorganic, nonmetallic solid that is prepared from powdered materials, is fabricated into products through the application of heat, and displays such characteristic properties as hardness, strength, low electrical conductivity, and brittleness.”
- The word ceramic comes from the Greek word "keramikos", which means "pottery.”
- They are typically crystalline in nature and are compounds formed between metallic and nonmetallic elements such as aluminum and oxygen (alumina- Al_2O_3), calcium and oxygen (calcia - CaO), and silicon and nitrogen (silicon nitride- Si_3N_4).
- Depending on their method of formation, ceramics can be dense or lightweight.
- Typically, they will demonstrate excellent strength and hardness properties; however, they are often brittle in nature.
- Ceramics can also be formed to serve as electrically conductive materials or insulators.
- Some ceramics, like superconductors, also display magnetic properties.
- They are also more resistant to high temperatures and harsh environments than metals and polymers.
- Due to ceramic materials wide range of properties, they are used for a multitude of applications.

Typical Properties Of Ceramics

- ✓ High hardness
- ✓ High elastic modulus
- ✓ Low ductility
- ✓ High dimensional stability
- ✓ Good wear resistance
- ✓ High resistance to corrosion and chemical attack
- ✓ High weather resistance
- ✓ High melting point
- ✓ High working temperature
- ✓ Low thermal expansion
- ✓ Low to medium thermal conductivity
- ✓ Good electrical insulation
- ✓ Low to medium tensile strength
- ✓ High compressive strength
- ✓ Medium machinability
- ✓ Opacity
- ✓ Brittleness
- ✓ Poor impact strength
- ✓ Low thermal shock resistance

Advantages And Disadvantages Of Ceramics

❖ Advantages of ceramics

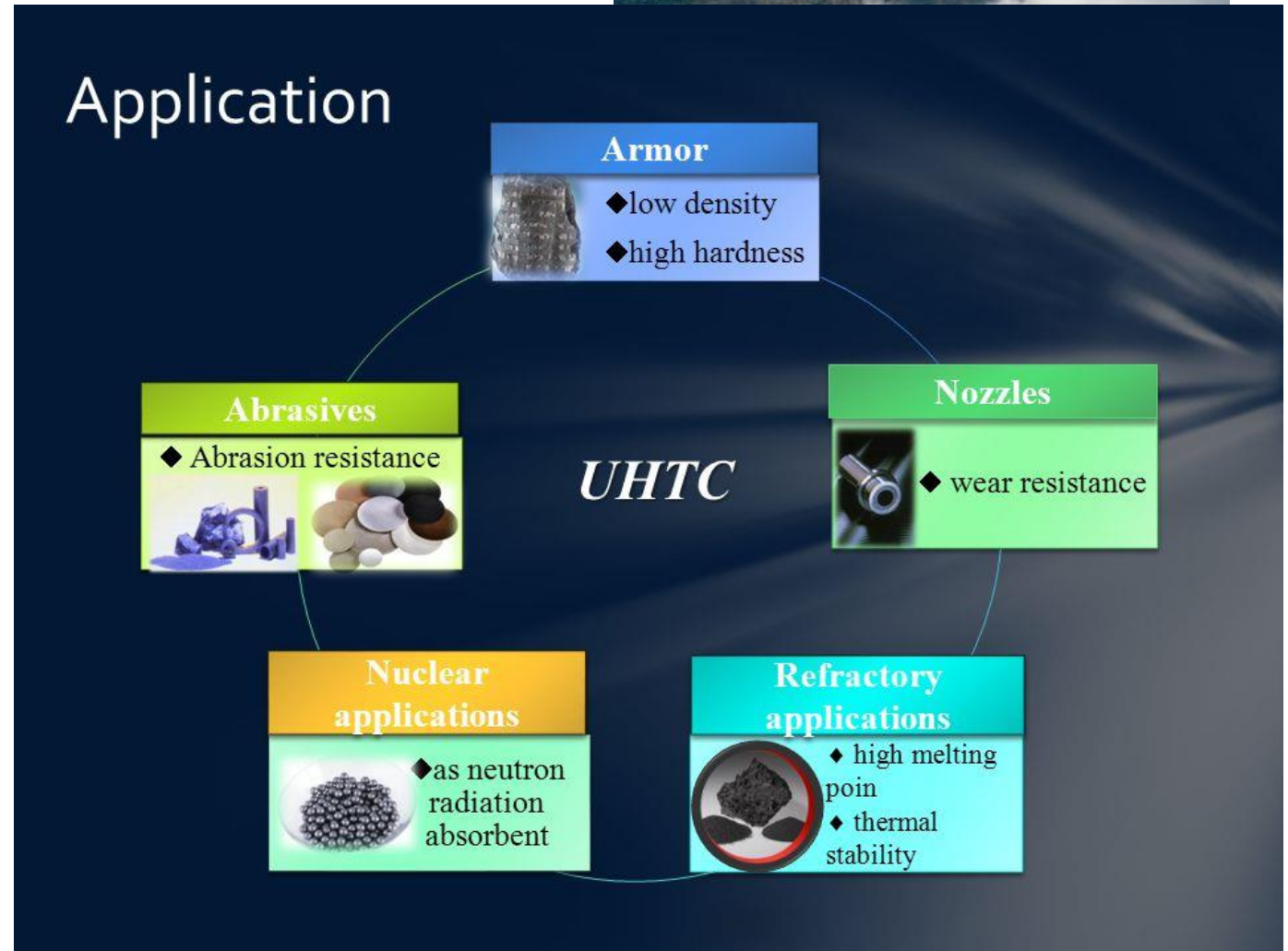
- ✓ Most of them have high hardness hence they are used as abrasive powder and cutting tools
- ✓ They have high melting point which makes them excellent refractory material
- ✓ They are good thermal insulators this is another reason to use them as refractory material
- ✓ They are high electric resistivity which makes them suitable to be used as an insulator
- ✓ They have low mass density which results in lightweight components
- ✓ They are generally chemically inert which makes them durable

❖ Disadvantages of ceramics

- ✓ They are brittle in nature
- ✓ They have almost zero ductility
- ✓ They have poor tensile strength
- ✓ They show a wide range in the variation of strength, even for the identical specimens
- ✓ They are difficult to shape and machine

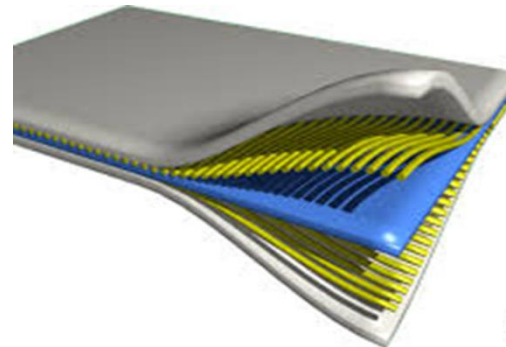
Applications Of Ceramics

- They are used in space industry because of their low weight
- They are used as cutting tools
- They are used as refractory materials
- They are used as thermal insulator
- They are used as electrical insulator



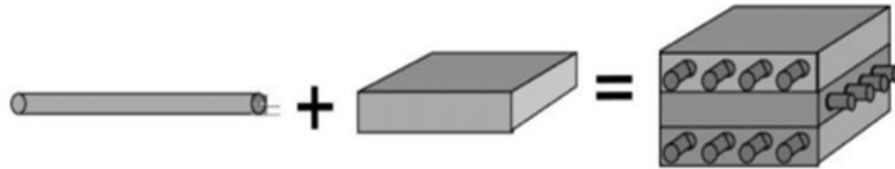
Composites

- A composite is commonly defined as a combination of two or more distinct materials, each of which retains its own distinctive properties, to create a new material with properties that cannot be achieved by any of the components acting alone.
- A number of composite materials have been engineered that consist of more than one material type.
- A composite is designed to display a combination of the best characteristics of each of the component materials.
- Fiberglass is a familiar example, in which glass fibers are embedded within a polymeric material.
- Fiberglass acquires strength from the glass and flexibility from the polymer. Many of the recent material developments have involved composite materials.



Composites

- Composite materials are said to have two phases.
- The reinforcing phase is the fibers, sheets, or particles that are embedded in the matrix phase.
- The reinforcing material and the matrix material can be metal, ceramic, or polymer.
- Typically, reinforcing materials are strong with low densities while the matrix is usually a ductile, or tough, material.



Fiber/Filament Reinforcement

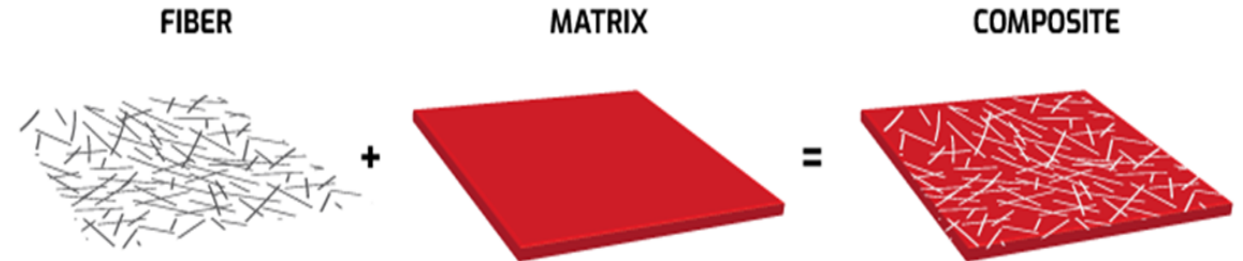
- High strength
- High stiffness
- Low density

Matrix

- Good shear properties
- Low density

Composite

- High strength
- High stiffness
- Good shear properties
- Low density



Composites

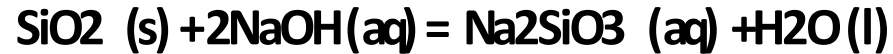
❖ **Some of the common classifications of composites are:**

- ✓ **Reinforced plastics**
- ✓ **Metal-matrix composites**
- ✓ **Ceramic-matrix composites**
- ✓ **Sandwich structures**
- ✓ **Concrete**

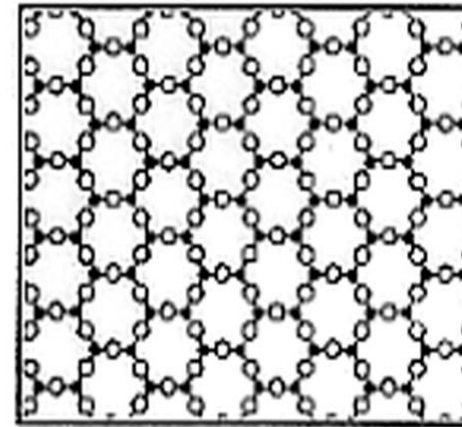
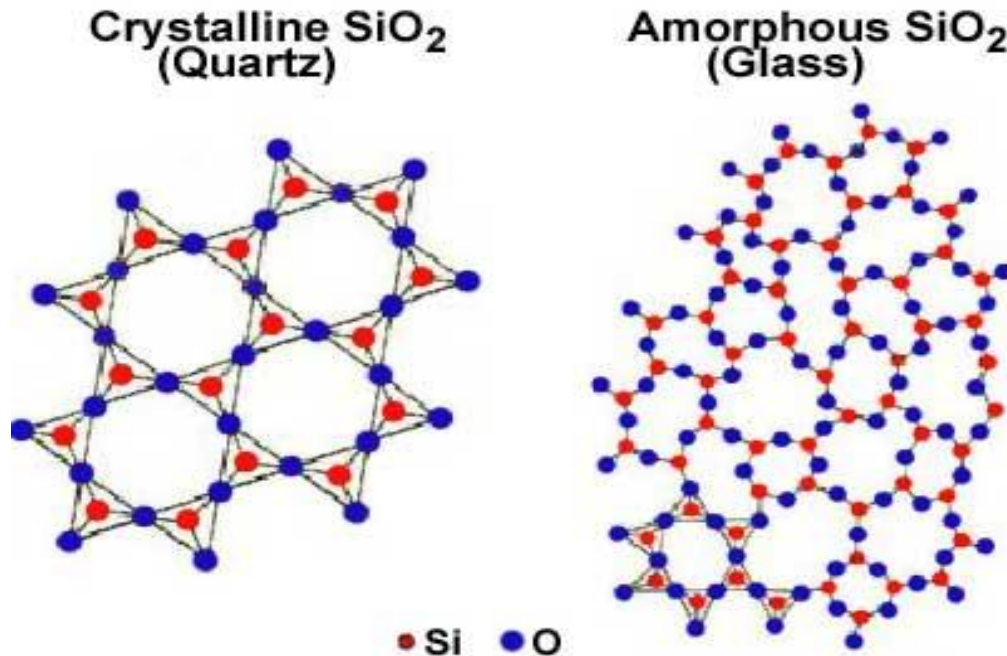
❖ **If the composite is designed and fabricated correctly, it combines the strength of the reinforcement with the toughness of the matrix to achieve a combination of desirable properties not available in any single conventional material.**

Glass

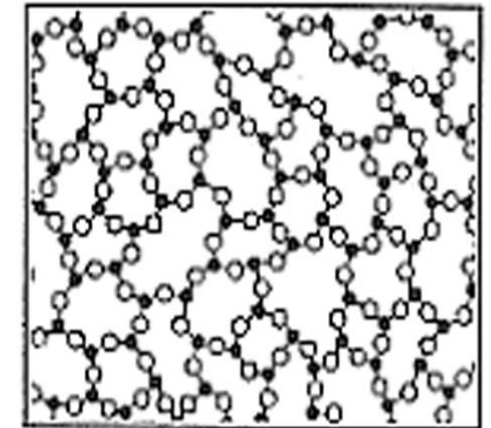
- When silicon oxide is heated to its melting point , the Si-O bonds break and it melts to form a viscous liquid. When cooled rapidly, silicate glass is formed.
- The silicon and oxygen atoms in glass are not arranged in a regular manner . This makes the glass an amorphous solid.
- Glass is soluble in hot obviously , concentrated alkalis as it contains silicon dioxide which is acidic.



- Hence, hot and concentrated alkalis should not be stored in glass containers.



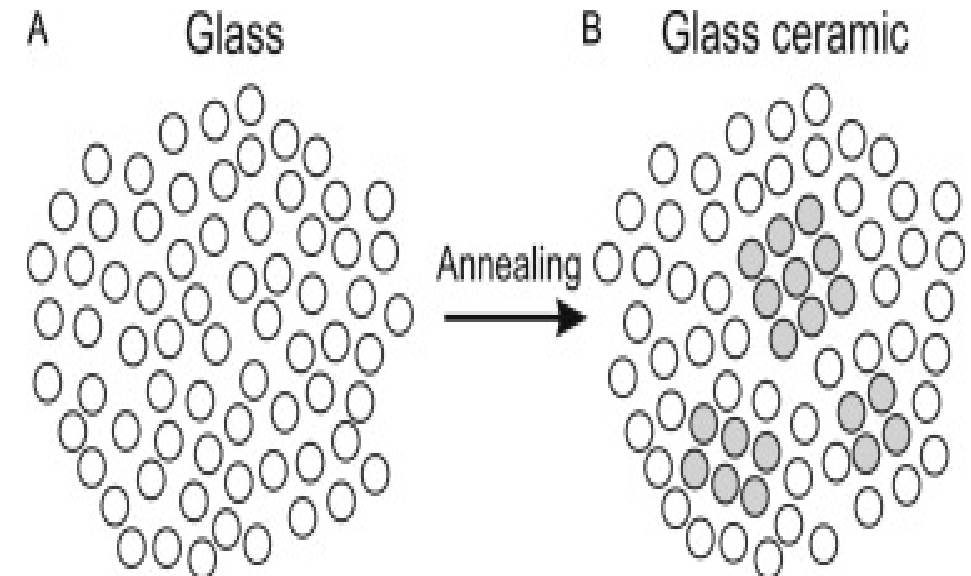
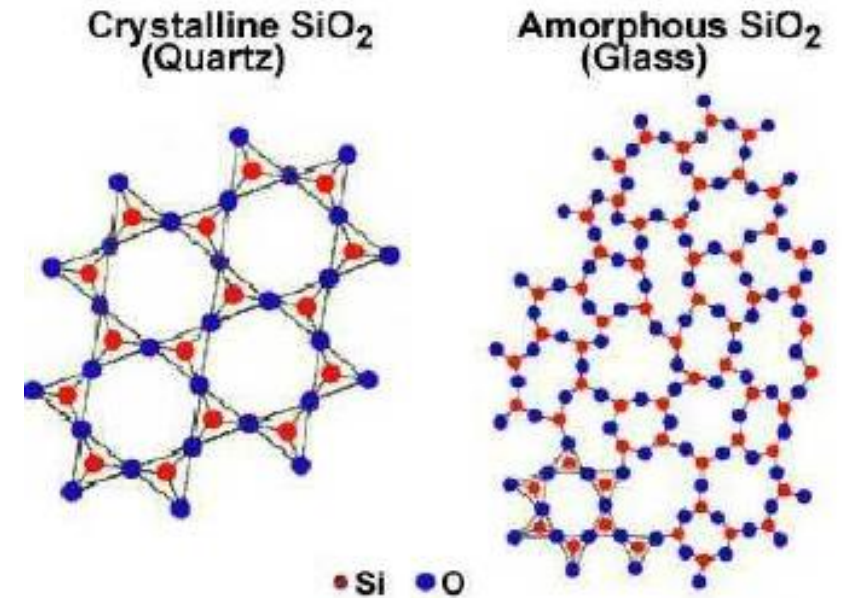
quartz



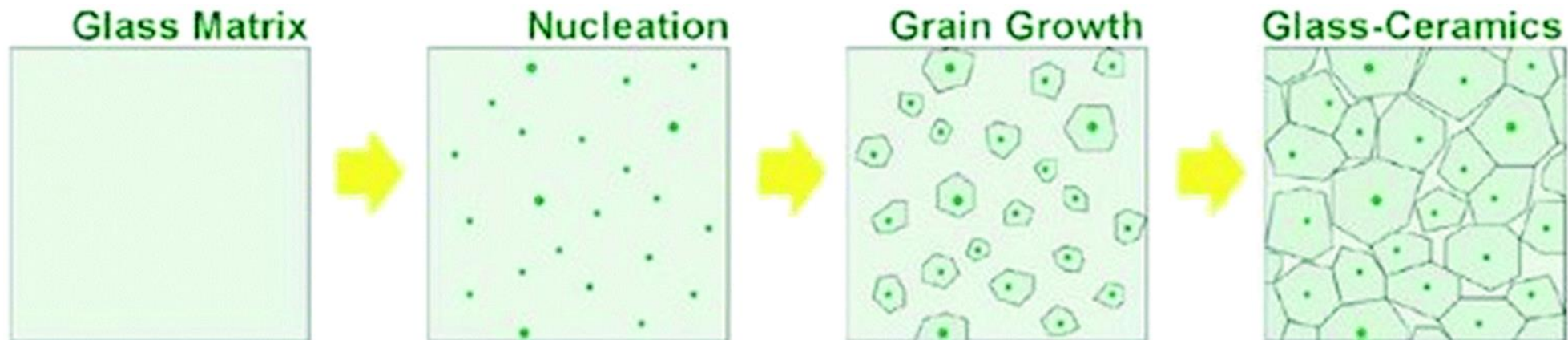
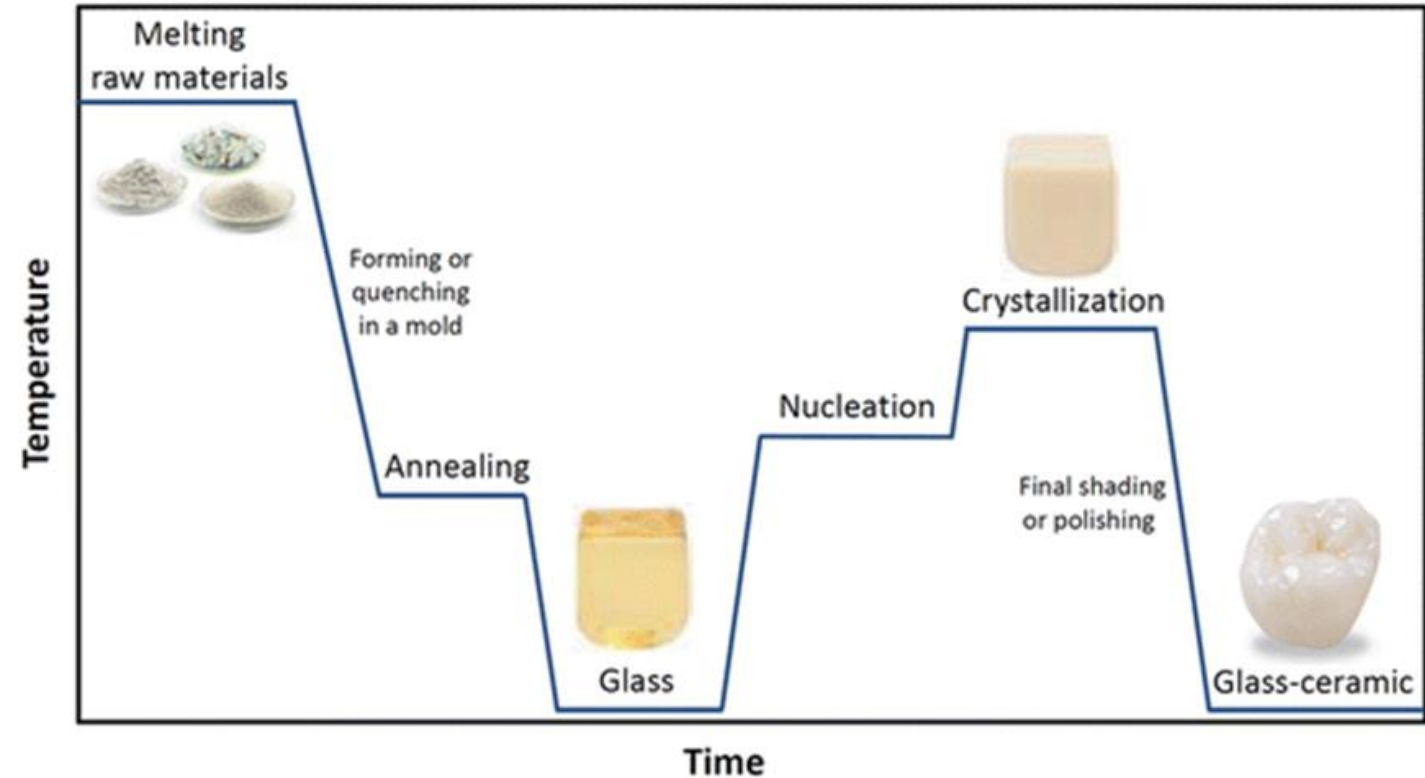
glass

Glass Ceramics

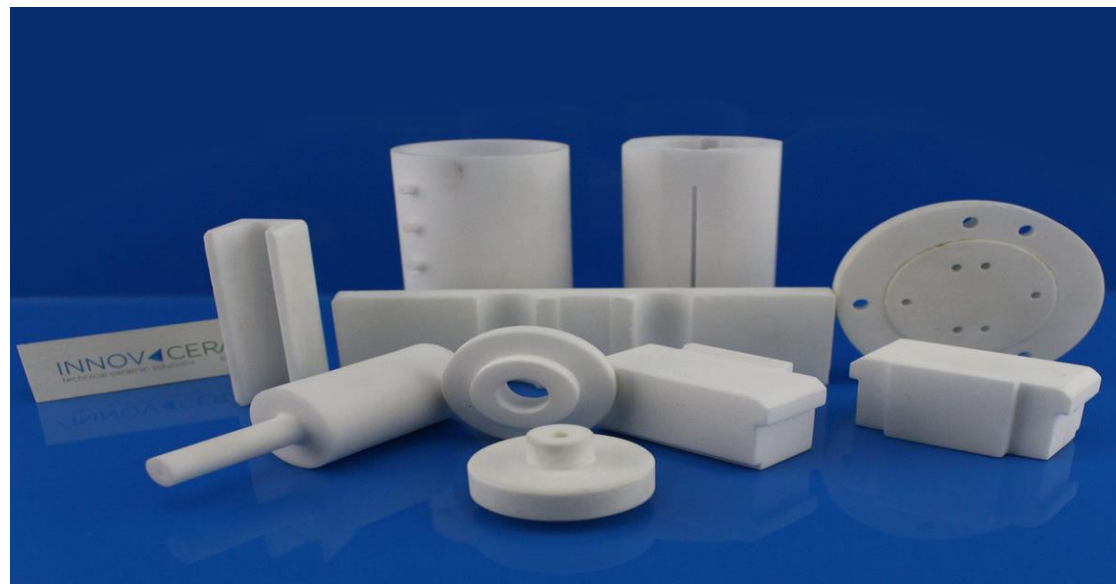
- Glass-ceramics are polycrystalline materials produced through controlled crystallization of base glass.
- Glass-ceramic materials share many properties with both glasses and ceramics.
- Glass-ceramics have an amorphous phase and one or more crystalline phases and are produced by a so-called "controlled crystallization".
- Glass-ceramics have the fabrication advantage of glass, as well as special properties of ceramics.
- When used for sealing, some glass-ceramics do not require brazing but can withstand brazing temperatures up to 700 °C.



Glass Ceramics



Glass Ceramics



Glass Ceramics

- Glass-ceramics usually have between 30% [mass/mass] and 90% [m/m] crystallinity and yield an array of materials with interesting properties like zero porosity, high strength, toughness, translucency or opacity, pigmentation, opalescence, low or even negative thermal expansion, high temperature stability, fluorescence, machinability, ferromagnetism, resorbability or high chemical durability, biocompatibility, bioactivity, ion conductivity, superconductivity, isolation capabilities, low dielectric constant, high resistivity and break-down voltage (The breakdown voltage of an insulator is the minimum voltage that causes a portion of an insulator to become electrically conductive.).
- These properties can be tailored by controlling the base-glass composition and by controlled heat treatment/crystallization of base glass.



Glass Ceramics

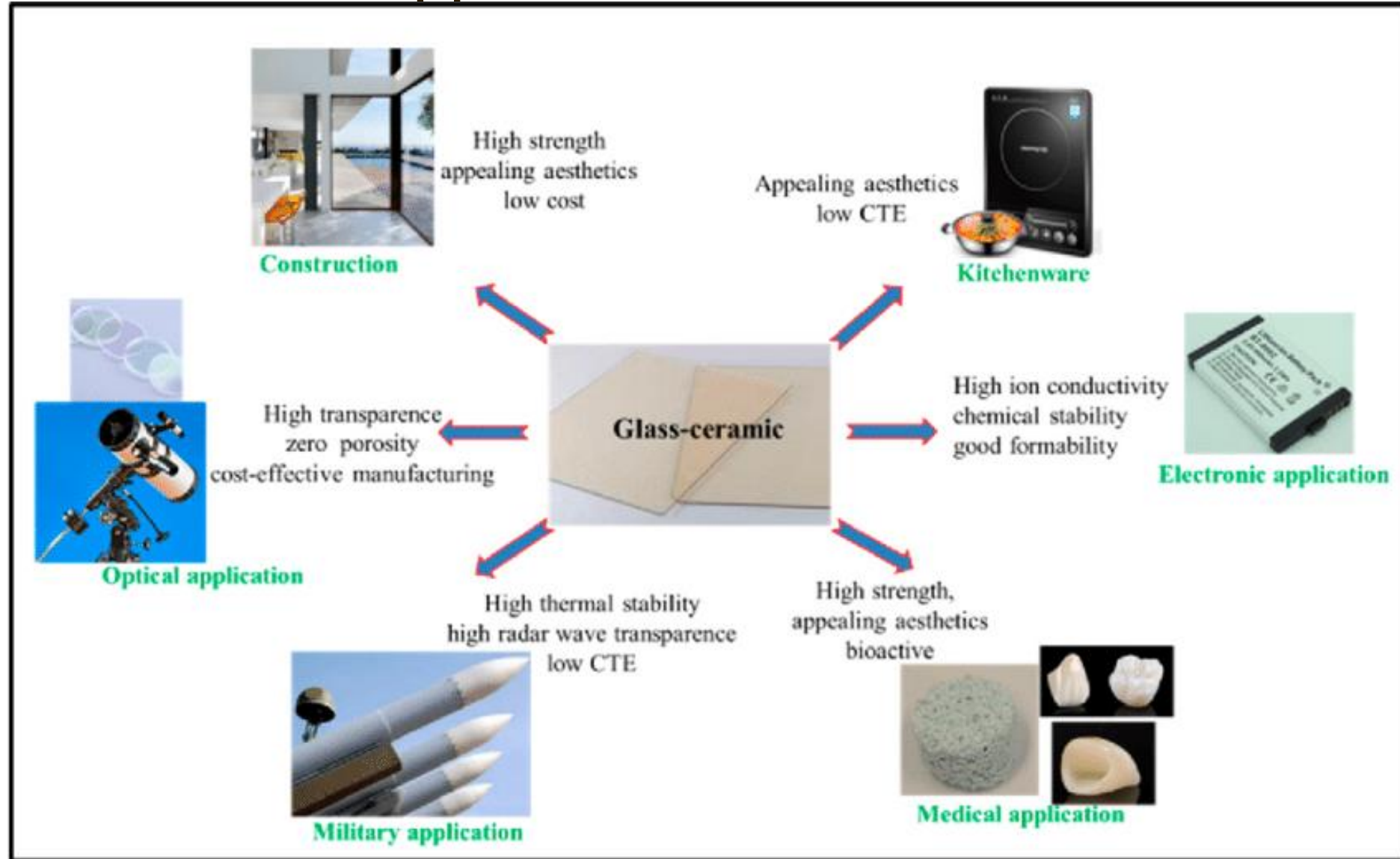
- In manufacturing, glass-ceramics are valued for having the strength of ceramic but the hermetic sealing (A hermetic seal is any type of sealing that makes a given object airtight (preventing the passage of air, oxygen, or other gases) properties of glass.
- Glass-ceramics are mostly produced in two steps.
- First, a glass is formed by a glass-manufacturing process.
- The glass is cooled down and is then reheated in a second step.
- In this heat treatment the glass partly crystallizes.
- In most cases nucleation agents are added to the base composition of the glass-ceramic.
- These nucleation agents aid and control the crystallization process.
- Because there is usually no pressing and sintering, glass-ceramics have no pores, unlike sintered ceramics.
- A wide variety of glass-ceramic systems exist, e.g., the $\text{Li}_2\text{O} \times \text{Al}_2\text{O}_3 \times n\text{SiO}_2$ system (LAS system), the $\text{MgO} \times \text{Al}_2\text{O}_3 \times n\text{SiO}_2$ system (MAS system), the $\text{ZnO} \times \text{Al}_2\text{O}_3 \times n\text{SiO}_2$ system (ZAS system).

Glass Ceramics- Applications

- With their thermal and mechanical characteristics, glass-ceramics have a range of applications, including:
- Cooktops (which you'll find in most modern kitchens)
- Household appliances including toasters and clothes irons
- Grills and BBQs
- Smartphone screens
- Infrared applications, such as in infrared heating elements
- In high-temperature furnaces as an insulation material, particularly due to their high thermal robustness.
- Biomedical engineering
- Advanced optics, for example for thermal shock resistant colour filters.



Glass Ceramics- Applications



Tempered Glass

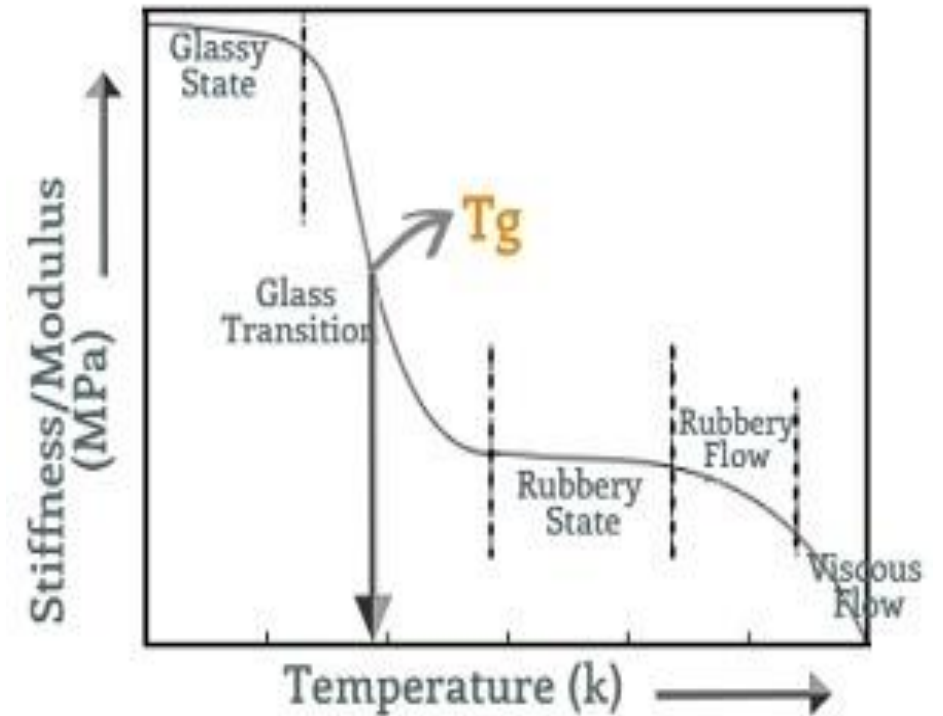
- Tempered or toughened glass is a type of safety glass processed by controlled thermal or chemical treatments to increase its strength compared with normal glass.
- Tempering puts the outer surfaces into compression and the interior into tension.
- Such stresses cause the glass, when broken, to shatter into small granular chunks instead of splintering into jagged shards as ordinary annealed glass does.
- The granular chunks are less likely to cause injury.
- **Tempered glass is used for its safety and strength in a variety of applications, including passenger vehicle windows, shower doors, aquariums, architectural glass doors and tables, refrigerator trays, mobile phone screen protectors, bulletproof glass components, diving masks, and plates and cookware.**





Glass Transition Temperature

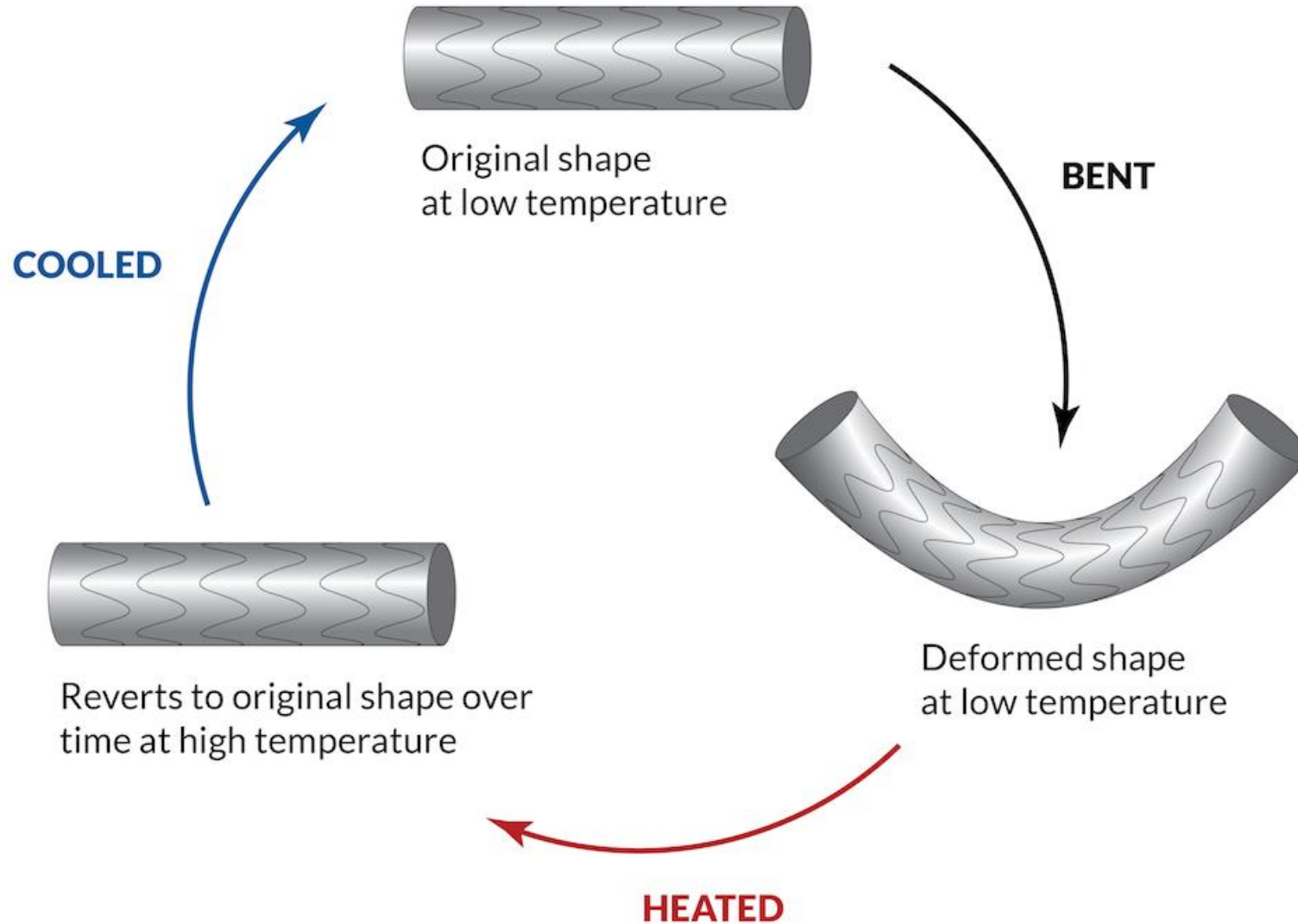
- The glass–liquid transition, or glass transition, is the gradual and reversible transition in amorphous materials (or in amorphous regions within semicrystalline materials) from a hard and relatively brittle "glassy" state into a viscous or rubbery state as the temperature is increased.
- An amorphous solid that exhibits a glass transition is called a glass.
- The reverse transition, achieved by supercooling a viscous liquid into the glass state, is called vitrification.
- The glass-transition temperature T_g of a material characterizes the range of temperatures over which this glass transition occurs. It is always lower than the melting temperature, T_m , of the crystalline state of the material, if one exists.
- The main difference between glass transition temperature and melting temperature is that glass transition temperature describes the transition of a glass state into a rubbery state whereas melting temperature describes the transition of a solid phase into a liquid phase.

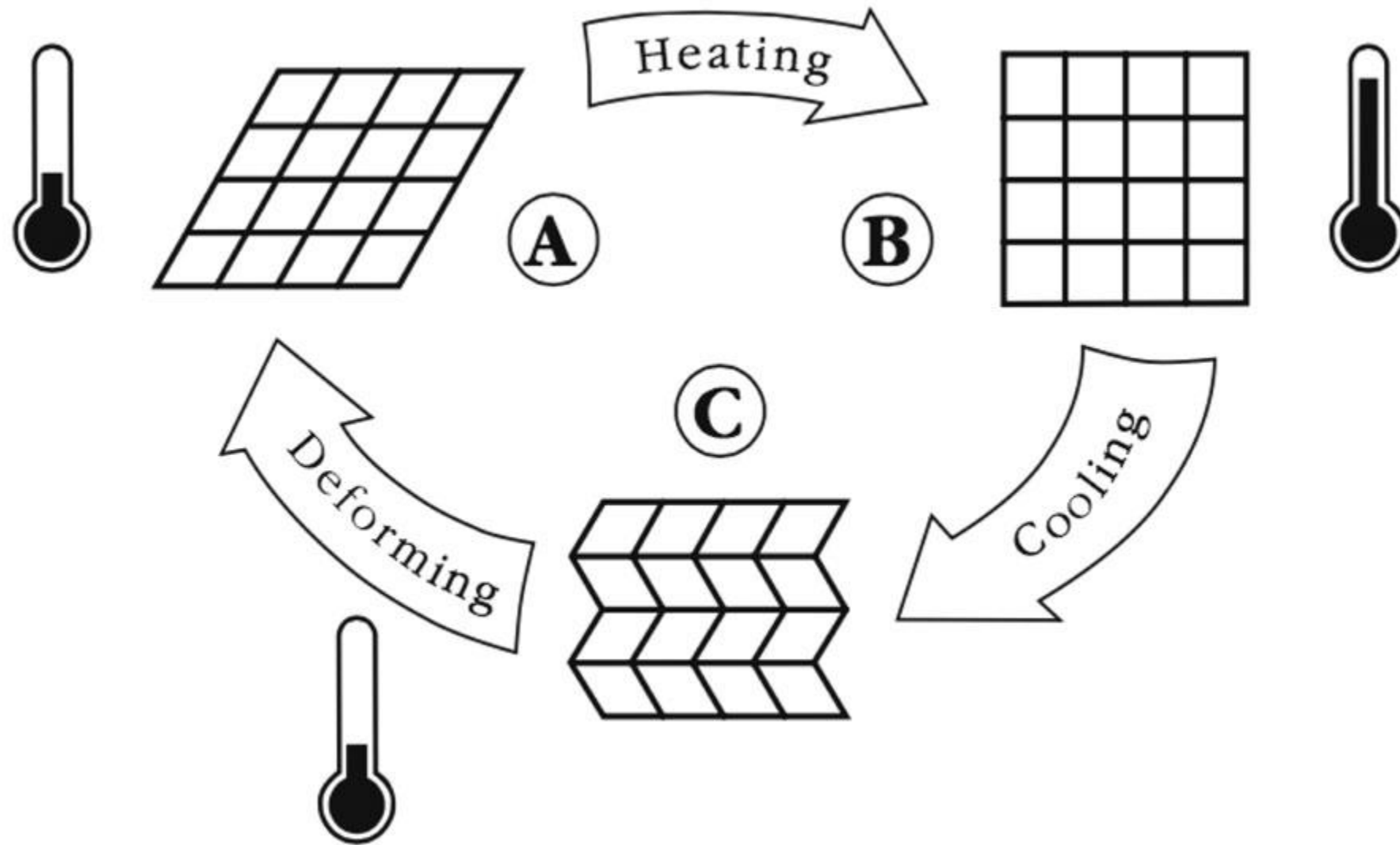


Shape memory alloys

- A shape-memory alloy (SMA, smart metal, memory metal, memory alloy, muscle wire, and smart alloy) is an alloy that “remembers” its original, cold-forged shape by returning to that pre-deformed shape when heated.
- Shape memory alloys (SMA) are materials that can undergo large pseudoelastic deformations while having the ability to recover to their original shape once subjected to a memory stimuli such as a specific temperature, stress, or strain.
- This material is an extremely lightweight, solid-state alternative to conventional actuators such as hydraulic, pneumatic, and motor-based systems.
- SMAs have applications in industries, including medical and aerospace.
- In most SMAs, a temperature change of only about 10°C is necessary to initiate this phase change.
- The three main types of SMAs are the copper-zinc-aluminum-nickel, copper-aluminum-nickel, and nickel-titanium (NiTi) alloys.

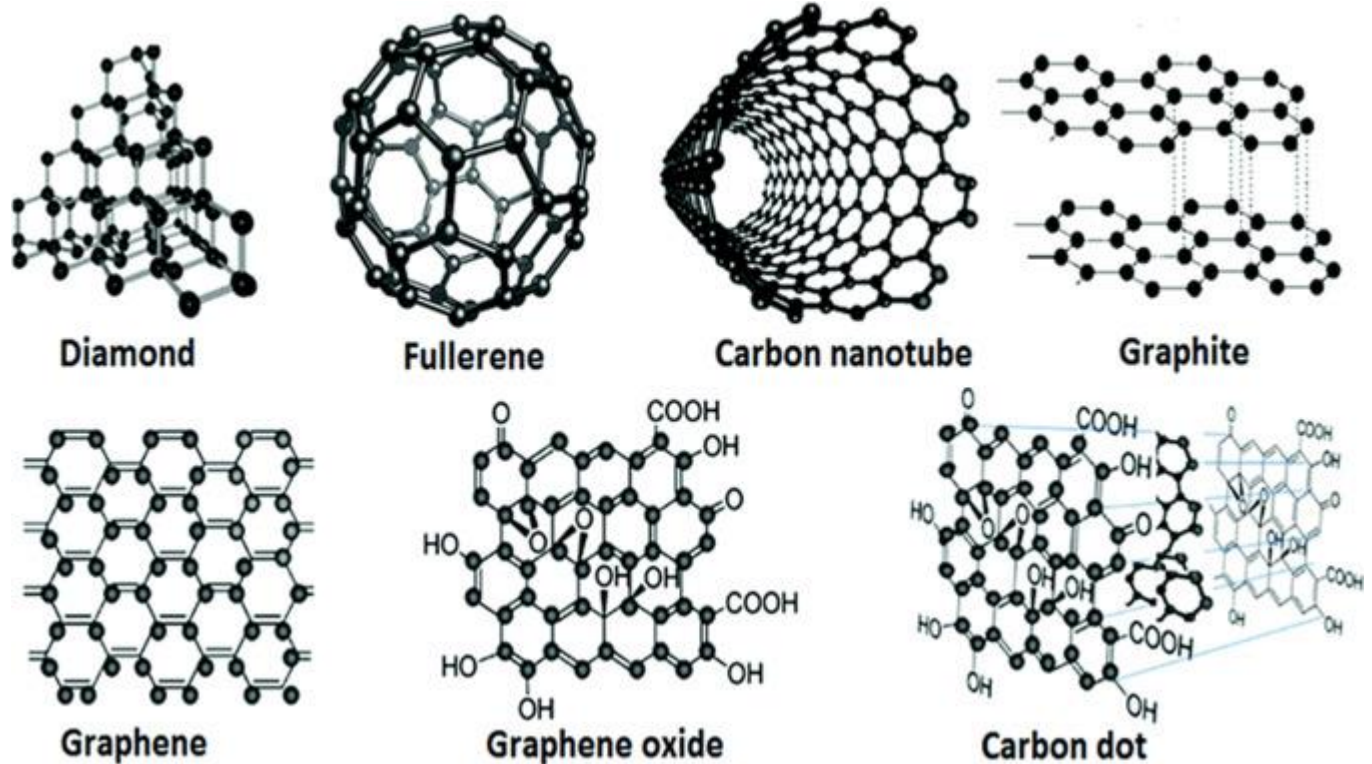
The Phase Transformation Process for SMAs





Nano Materials

- Nanomaterials are defined as materials that measure on the nanoscale (1–100 nm) or materials that contain nanoscale structures internally or on their surfaces.
- 'A natural, incidental or manufactured material containing particles, in an unbound state or as an aggregate or as an agglomerate and where, for 50 % or more of the particles in the number size distribution, one or more external dimensions is in the size range 1 nm–100 nm.'



Applications

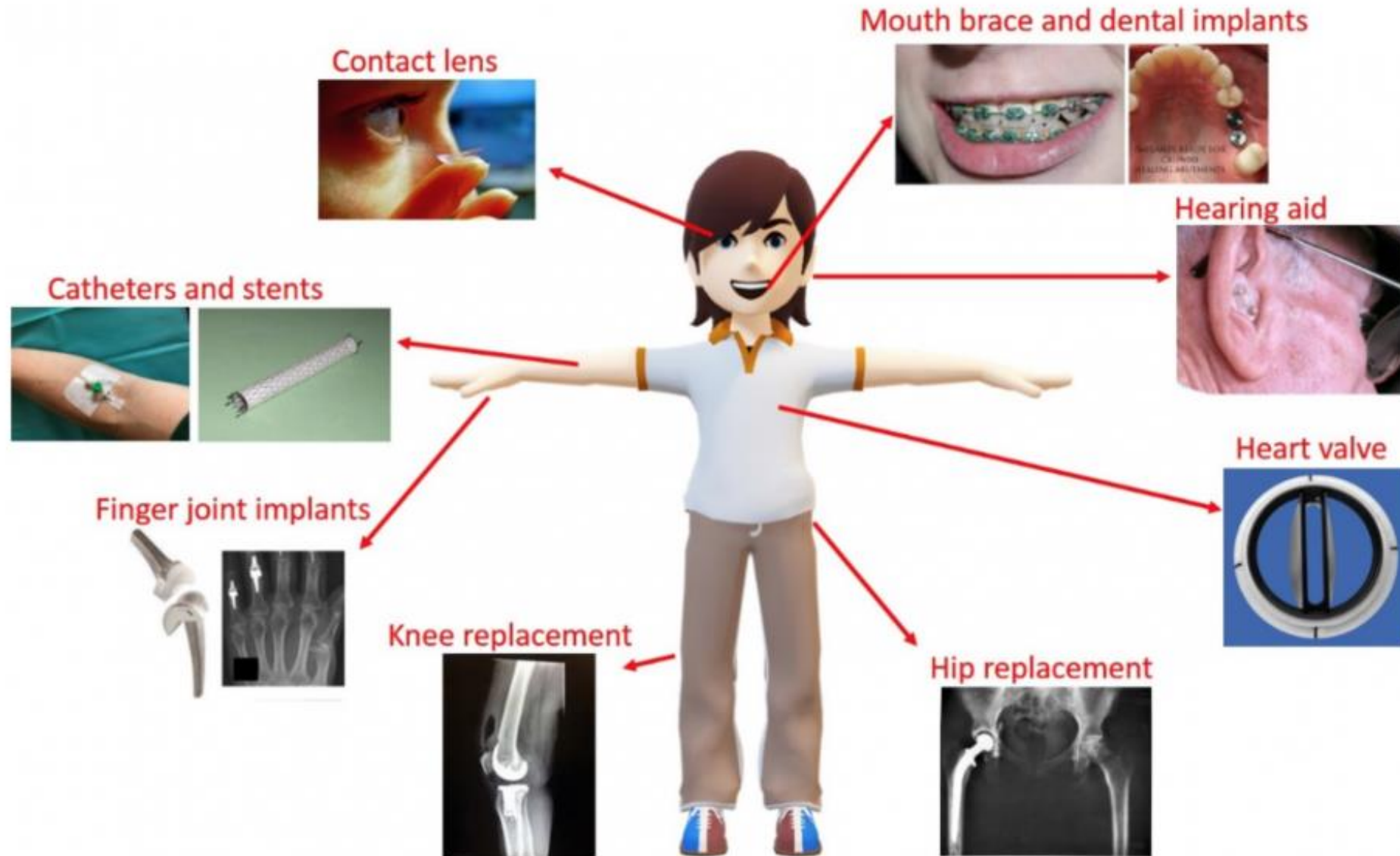
- The healthcare field, utilises nanomaterials in various ways, with one major use being drug delivery.
- One example of this process is whereby nanoparticles are being developed to assist the transportation of chemotherapy drugs directly to cancerous growths, as well as to deliver drugs to areas of arteries that are damaged to fight cardiovascular disease.
- Carbon nanotubes are also being developed to be used in processes such as the addition of antibodies to the nanotubes to create bacteria sensors.
- In aerospace, carbon nanotubes can be used in the morphing of aircraft wings.
- The nanotubes are used in a composite form to bend in response to the application of an electric voltage.
- Environmental preservation processes make use of nanomaterials mainly as nanowires. Applications are being developed to use the nanowires - zinc oxide nanowires- in flexible solar cells as well as to play a role in the treatment of polluted water.
- In the cosmetics industry, mineral nanoparticles –such as titanium oxide –are used in sunscreen.
- The sports industry has been producing baseball bats that have been made with carbon nanotubes.

BIOMATERIALS

- Biomaterials are employed in components implanted into the human body for replacement of diseased or damaged body parts.
- These materials must not produce toxic substances and must be compatible with body tissues (i.e., must not cause adverse biological reactions).
- All of the above materials (combinations - alloys) —metals, ceramics, polymers, composites, and semiconductors—may be used as biomaterials.



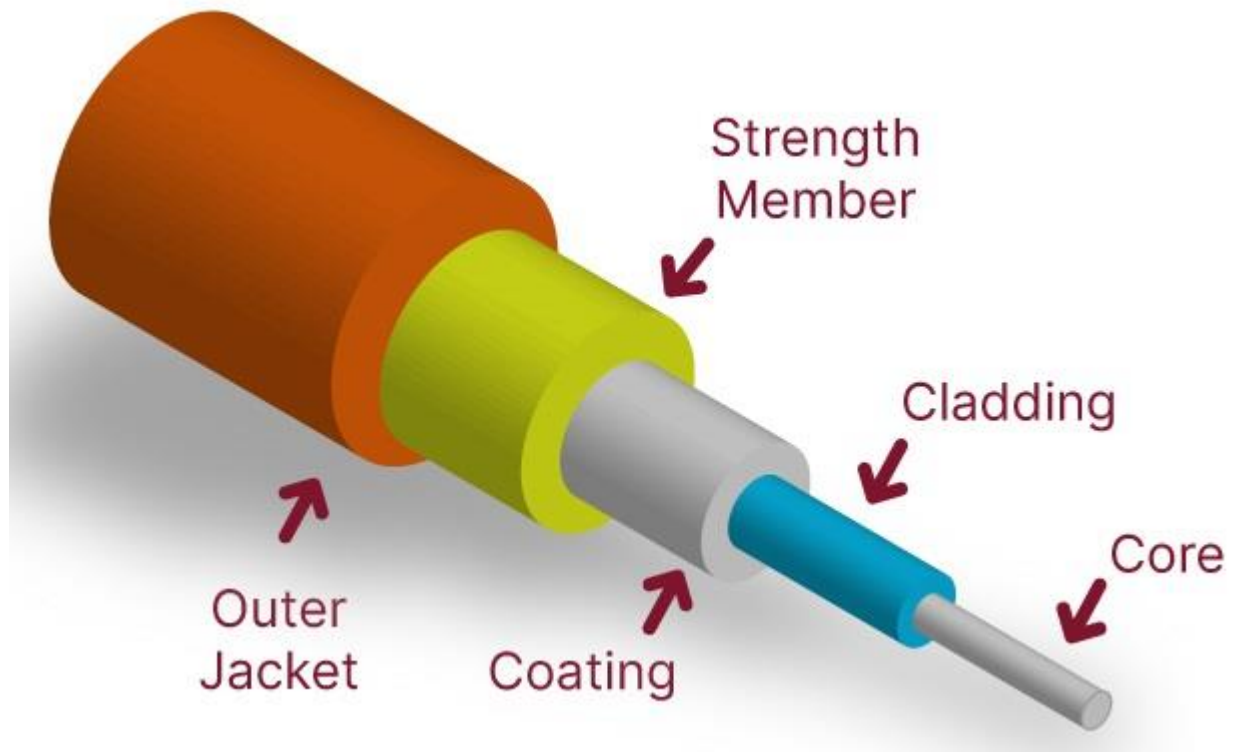
Biomaterials



Optical fiber

- Optical fibre is the technology associated with data transmission using light pulses travelling along with a long fibre which is usually made of plastic or glass.
- Metal wires are preferred for transmission in optical fibre communication as signals travel with fewer damages.
- Optical fibres are also unaffected by electromagnetic interference.
- The fibre optical cable uses the application of total internal reflection of light.
- The fibres are designed such that they facilitate the propagation of light along with the optical fibre depending on the requirement of power and distance of transmission.
- Single-mode fibre is used for long-distance transmission, while multimode fibre is used for shorter distances.
- The outer cladding of these fibres needs better protection than metal wires.

Optical fiber



Basic Operation of an Optical Fiber

